



National University of Computer and Emerging Sciences

Customer Personality Analysis Using Machine Learning

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Abstract

This report shows the use of machine learning methods on the Customer Personality Analysis dataset that was acquired on Kaggle. The project is concerned with two predictive analytics tasks, which are regression-based prediction of customer total spending behavior and classification-based prediction of customer campaign responses. Several machine learning models were trained, tested, and compared to determine the most effective methods of customer analytics.

Linear Regression, Support Vector Regression (SVR), Decision Tree Regressor, Random Forest Regressor, and K-Nearest Neighbors (KNN) Regressor were used in regression analysis. The best predictive performance was recorded by the Random Forest Regressor which had the highest R² Score and minimum prediction errors.

To perform classification analysis, the following types of classifiers were used: Logistic Regression, KNN Classifier, Decision Tree Classifier, Random Forest Classifier, and Support Vector Machine (SVM)

Classifier to predict customer marketing campaign responses. The best overall classification performance was obtained with the Random Forest Classifier.

The project shows how machine learning can be used to assist in customer segmentation, marketing optimization, customer behavior prediction, and business decision-making through predictive analytics methods.

1. Introduction

Contemporary companies gather extensive customer information in the form of transactions, online communications, and marketing efforts. The analysis of this information can assist organizations to understand customer behavior better, enhance marketing strategies and optimize business performance. Machine learning gives organizations the capability to detect latent trends in customer data and make correct predictions based on past data.

Machine learning has found a significant application in customer analytics since companies seek to determine valuable customers, forecast buying behavior, and enhance campaign targeting. Predictive models can assist companies in distributing marketing resources more effectively and enhance customer satisfaction.

This project uses the Customer Personality Analysis dataset to analyze customer demographic information, purchasing activity, and campaign responses. The project uses regression and classification machine learning methods. Customer total spending behavior is predicted using regression models and whether customers will respond positively to marketing campaigns is predicted using classification models.

The project adheres to a full machine learning process such as exploratory data analysis, feature engineering, preprocessing, model training, evaluation, and comparison. The project results are valuable in understanding customer behavior and illustrate the practicality of machine learning in marketing analytics.

2. Dataset

The dataset of this project is the Customer Personality Analysis dataset that was retrieved on Kaggle. The data set includes customer demographic data, buying behavior and campaign response data gathered to analyze the marketing.

The dataset falls under the marketing and customer analytics field and includes data on:

- Customer income
- Education level
- Marital status
- Purchase behavior

- Web activity
- Campaign responses
- Product spending categories

The initial dataset had customer records in various categories of spending such as wines, fruits, meat products, fish products, sweet products, and gold products.

The data has 2240 rows and various customer-related characteristics that are applied in predictive analytics.

memory usage: 587.6+ KB

	ID	Year_Birth	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	MntFruits	MntMeatProducts	...	Nu
count	2240.000000	2240.000000	2216.000000	2240.000000	2240.000000	2240	2240.000000	2240.000000	2240.000000	2240.000000	...	Nu
mean	5592.159821	1968.805804	52247.251354	0.444196	0.506250	2013-07-10 10:01:42.857142784	49.109375	303.935714	26.302232	166.950000	...	Nu
min	0.000000	1893.000000	1730.000000	0.000000	0.000000	2012-07-30 00:00:00	0.000000	0.000000	0.000000	0.000000	...	Nu
25%	2828.250000	1959.000000	35303.000000	0.000000	0.000000	2013-01-16 00:00:00	24.000000	23.750000	1.000000	16.000000	...	Nu
50%	5458.500000	1970.000000	51381.500000	0.000000	0.000000	2013-07-08 12:00:00	49.000000	173.500000	8.000000	67.000000	...	Nu
75%	8427.750000	1977.000000	68522.000000	1.000000	1.000000	2013-12-30 06:00:00	74.000000	504.250000	33.000000	232.000000	...	Nu
max	11191.000000	1996.000000	666666.000000	2.000000	2.000000	2014-06-29 00:00:00	99.000000	1493.000000	199.000000	1725.000000	...	Nu
std	3246.662198	11.984069	25173.076661	0.538398	0.544538	NaN	28.962453	336.597393	39.773434	225.715373	...	Nu

8 rows x 27 columns

2.1 Regression Target Variable

The regression target variable that will be used in this project is Total_Spending. This variable was developed by summing the customer spending in all the product categories. Total_Spending is the total amount of money that each customer contributes.

The target variable is significant as customer spending behavior has a direct impact on business revenue generation, customer value analysis, and marketing strategy planning.

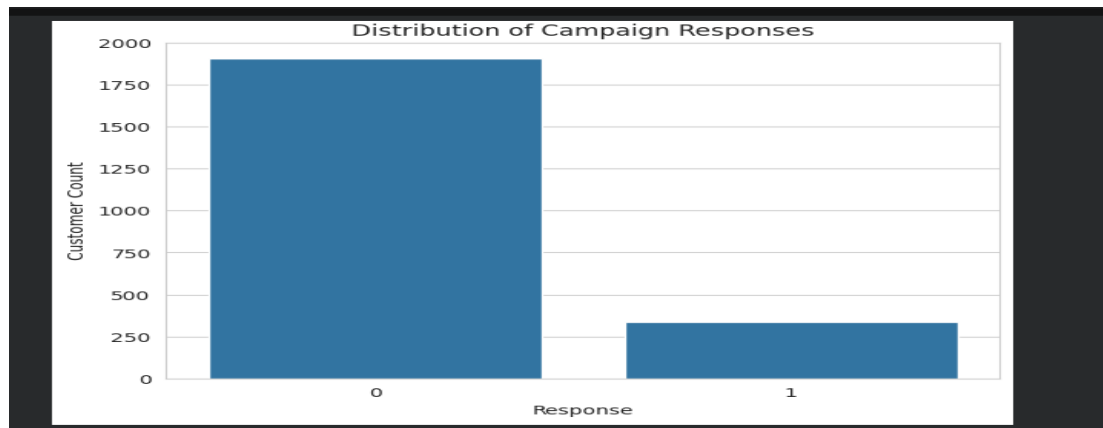
2.2 Classification Target Variable

Response is the classification target variable that will be used in this project.

Value 1 means that the customer accepted the marketing campaign.

Value 0 means that the customer did not react well to the campaign.

The behavior of customers in response to the campaign is important to predict since the business wants to know which customers are more likely to respond to the marketing campaign.



2.3 Data Quality and Preprocessing Summary.

Data quality analysis revealed that the dataset had few missing values in the Income column. Median imputation was used to fill in these missing values to maintain the structure of the dataset.

There were also duplicate rows in the dataset. Despite the presence of duplicates, the dataset was still suitable to be used in predictive analysis.

Categorical variables like Education and Marital_Status were coded into numerical values to enable the machine learning algorithms to work with the data.

StandardScaler was used to perform feature scaling to normalize the values of features and enhance the performance of the model.

An 80/20 split ratio was used to split the dataset into training and testing sets. Regression and classification were done using separate datasets.

To prevent target leakage, spending-related variables used to create the Total_Spending target variable were removed from regression model features.

```

Missing Values:
Year_Birth      0
Education       0
Marital_Status  0
Income         24
Kidhome        0
Teenhome       0
Dt_Customer     0
Recency        0
MntWines       0
MntFruits      0
MntMeatProducts 0
MntFishProducts 0
MntSweetProducts 0
MntGoldProds   0
NumDealsPurchases 0
NumWebPurchases 0
NumCatalogPurchases 0
NumStorePurchases 0
NumWebVisitsMonth 0
AcceptedCmp3    0
AcceptedCmp4    0
AcceptedCmp5    0
AcceptedCmp1    0
AcceptedCmp2    0
Complain       0
Response       0
Total_Spending  0
Age            0
Children       0
Total_Purchases 0
dtype: int64

Duplicate Rows:
182

```

```

Regression Training Shape: (1792, 21)
Regression Testing Shape: (448, 21)

Classification Training Shape: (1792, 27)
Classification Testing Shape: (448, 27)

```

3. Methodology

The project was based on a full machine learning process that included exploratory data analysis, feature selection, preprocessing, model training, and evaluation.

3.1 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to learn the customer behavior patterns, feature relationships, and campaign response distribution.

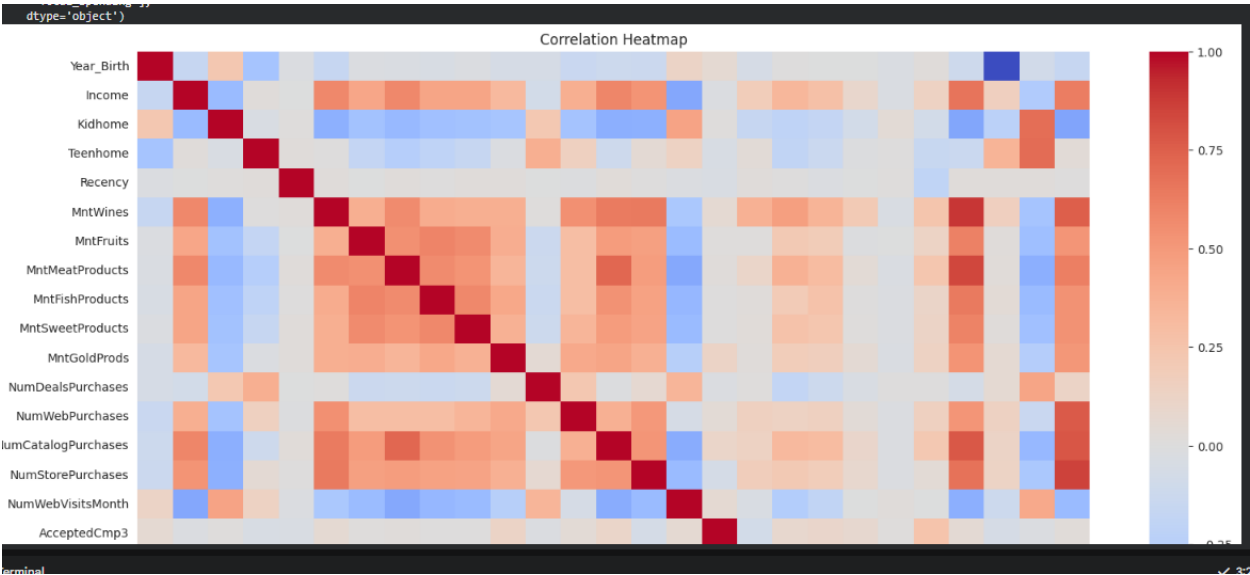
Distribution of Total Spending

The histogram of Total_Spending was positively skewed. The majority of customers had comparatively lower spending values, whereas a smaller number of customers had a significantly higher amount of spending.

This implies that there are high-value customers who can be significant in targeted marketing and customer retention strategies.

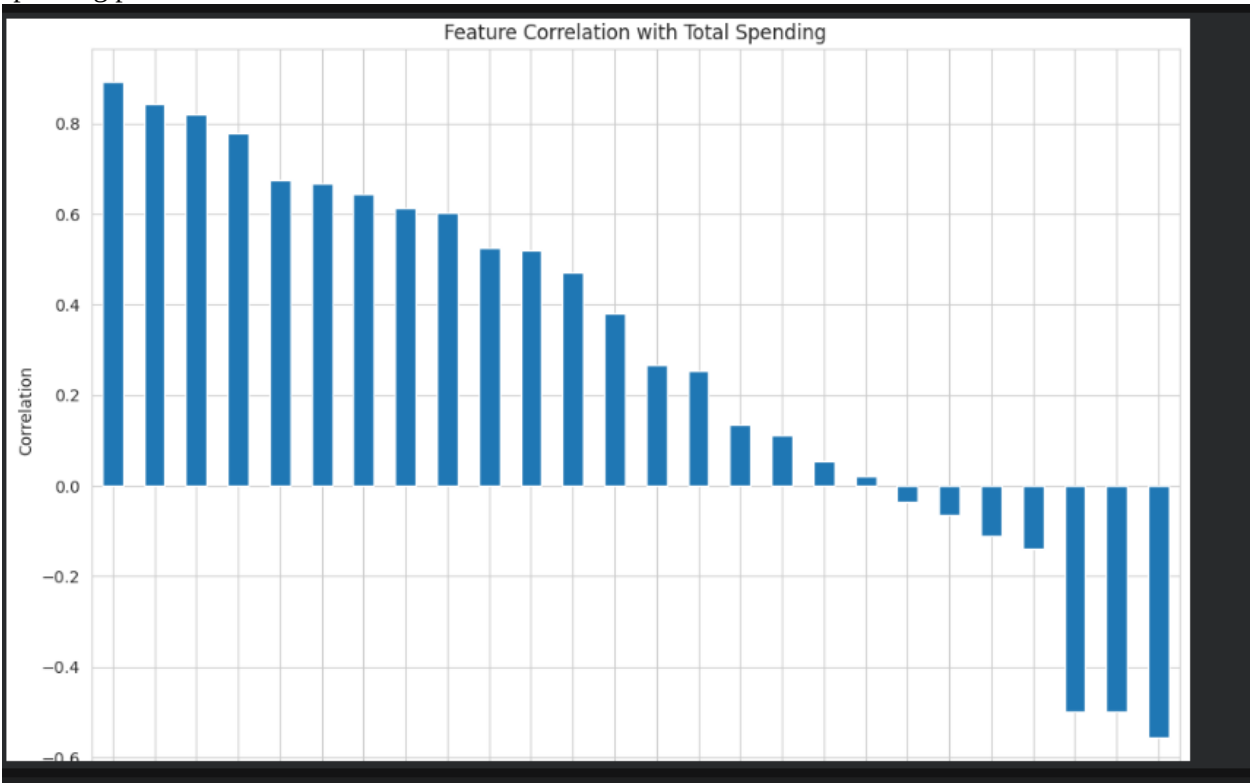
The campaign response countplot revealed that the majority of customers did not respond well to marketing campaigns. This shows imbalance of classes in the classification target variable.

The imbalance demonstrates the significance of applying evaluation metrics like Precision, Recall, and F1 Score rather than focusing on Accuracy.

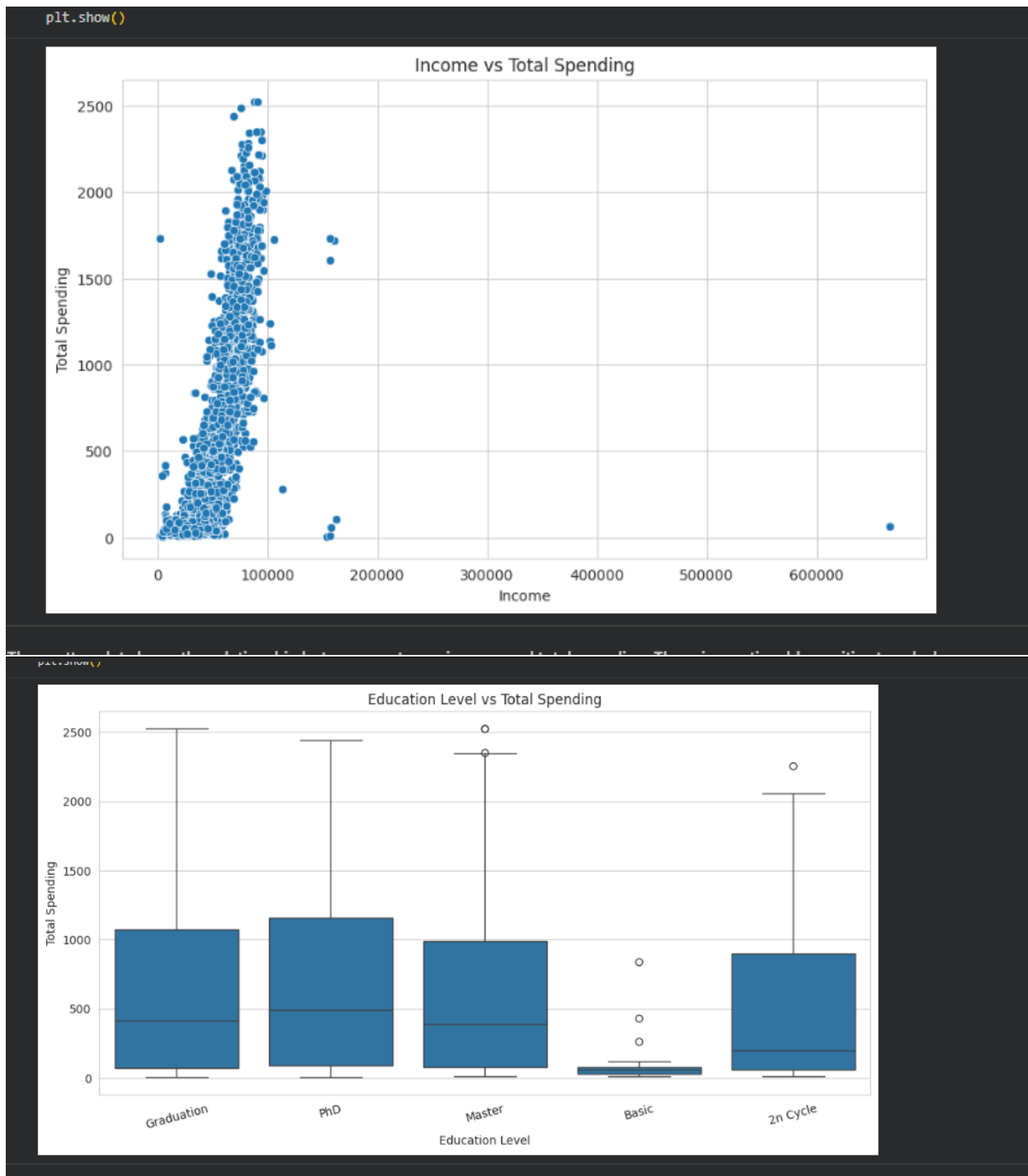


The correlation heatmap indicated that there were strong positive correlations between the variables related to spending and the Total_Spending target variable.

There were also significant correlations between income and purchasing behavior variables and customer spending patterns.



The scatter plot showed that there was a positive correlation between customer income and total spending behavior. The more affluent customers tended to spend more in general.



The boxplot analysis revealed that there were differences in spending behavior among the various levels of education. Some types of education had higher median spending values.

3.2 Feature Selection

Correlation analysis and business relevance were used to select the features.

Some of the redundant columns were eliminated since they were not meaningful predictors. Other characteristics like Age, Children, and Total_Purchases were also engineered to enhance predictive analysis.

Correlation analysis revealed that the variables associated with income, purchasing activity, and customer engagement were more strongly correlated with customer spending and campaign response behavior.

The process of feature selection enhanced the performance of the model and minimized the unwarranted complexity.

3.3 Preprocessing Pipeline

The preprocessing pipeline was composed of the following steps:

- Median imputation to deal with missing values.
- Duplicate row checking.
- Encoding categorical variables.
- Feature engineering.
- StandardScaler feature scaling.
- Feature-target separation.
- Train-test split with an 80/20 split.
- Prevention of target leakage by eliminating spending-related variables in regression features.

These preprocessing measures made sure that the data was well prepared to be modelled using machine learning.

3.4 Models Trained

Regression models and classification models were trained and tested (five each).

Regression Models:

- Linear Regression
- Support Vector Regression (SVR)
- Decision Tree Regressor
- Random Forest Regressor
- K-Nearest Neighbors (KNN) Regressor.

Classification Models:

- Logistic Regression
- KNN Classifier
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machine (SVM) Classifier.

Regression models were tested on:

- MAE
- MSE
- RMSE
- R² Score

The evaluation of classification models was based on:

- Accuracy
- Precision
- Recall
- F1 Score

4. Regression Results

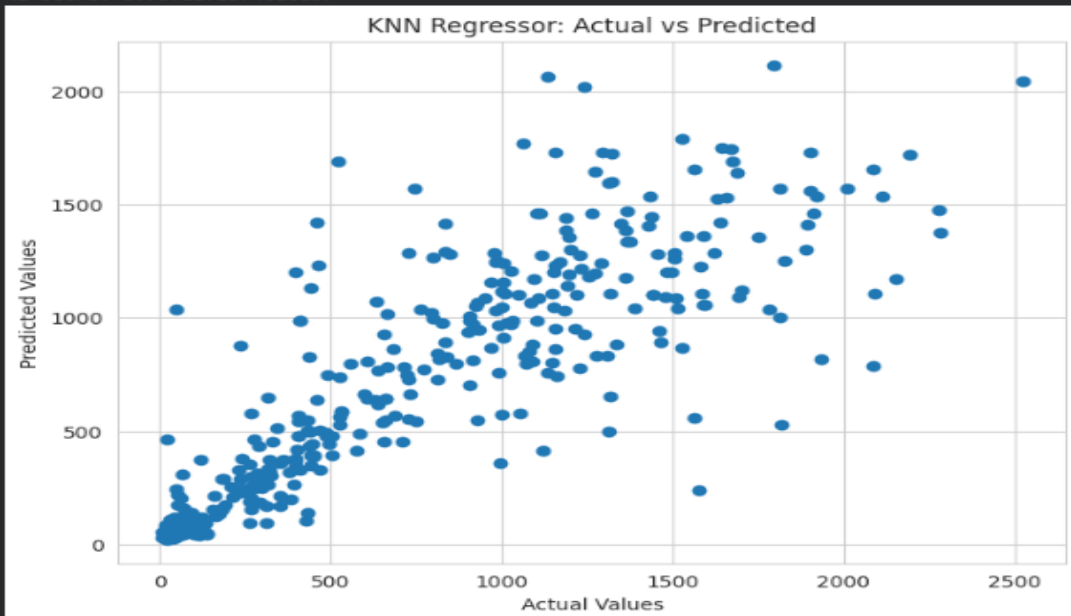
The Total_Spending target variable was used to predict customer spending behavior using multiple regression models.

Regression models AND comparison

***	Model	MAE	MSE	RMSE	R2 Score
0	Linear Regression	191.745944	86786.643806	294.595729	0.755919
1	SVR	385.890666	315723.156055	561.892477	0.112050
2	Decision Tree	142.127232	85346.899554	292.141917	0.759968
3	Random Forest	110.336696	42841.617633	206.982167	0.879511
4	KNN	172.083482	86356.916518	293.865474	0.757127

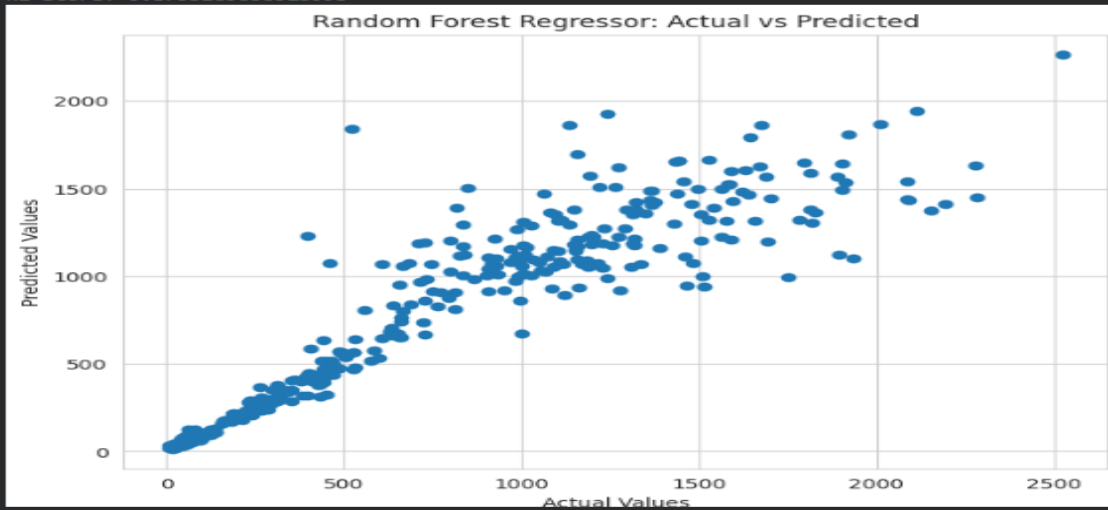
n:

KNN REGRESSOR RESULTS
MAE: 172.08348214285712
MSE: 86356.91651785713
RMSE: 293.8654735042161
R2 Score: 0.7571271374016267



Terminal

RANDOM FOREST REGRESSOR RESULTS
MAE: 110.33669642857141
MSE: 42841.61763348214
RMSE: 206.98216742869937
R2 Score: 0.8795109096925993



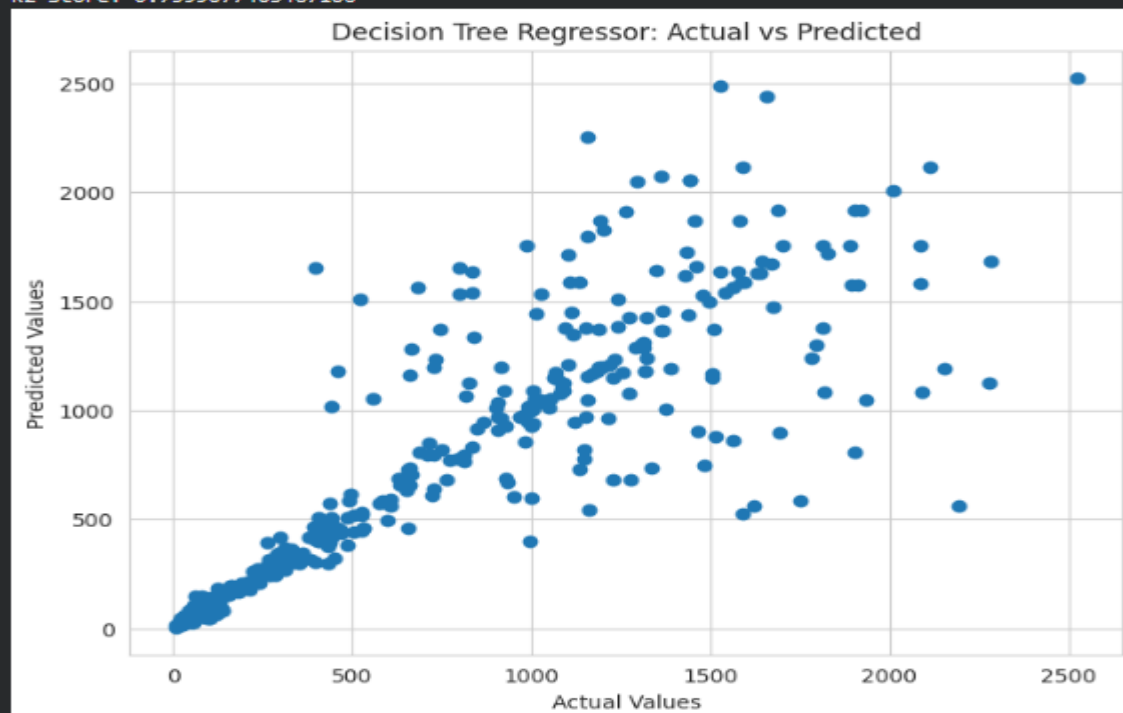
DECISION TREE REGRESSOR RESULTS

MAE: 142.12723214285714

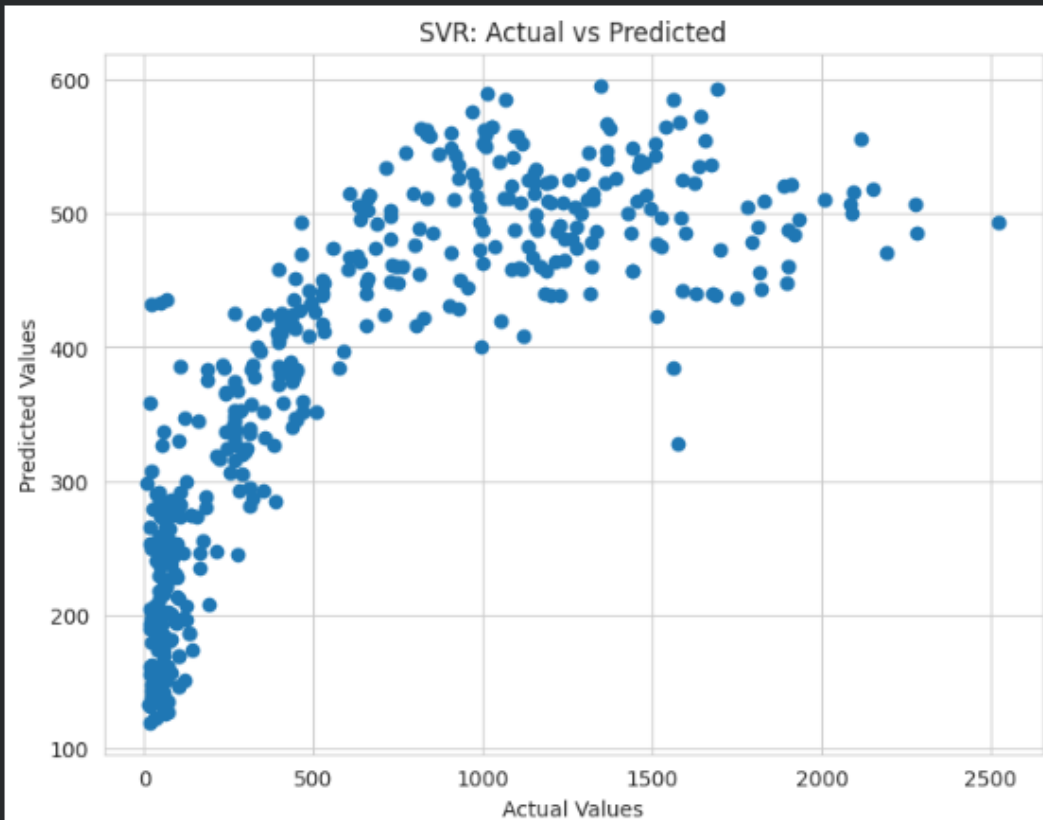
MSE: 85346.89955357143

RMSE: 292.14191680341156

R2 Score: 0.7599677403467106

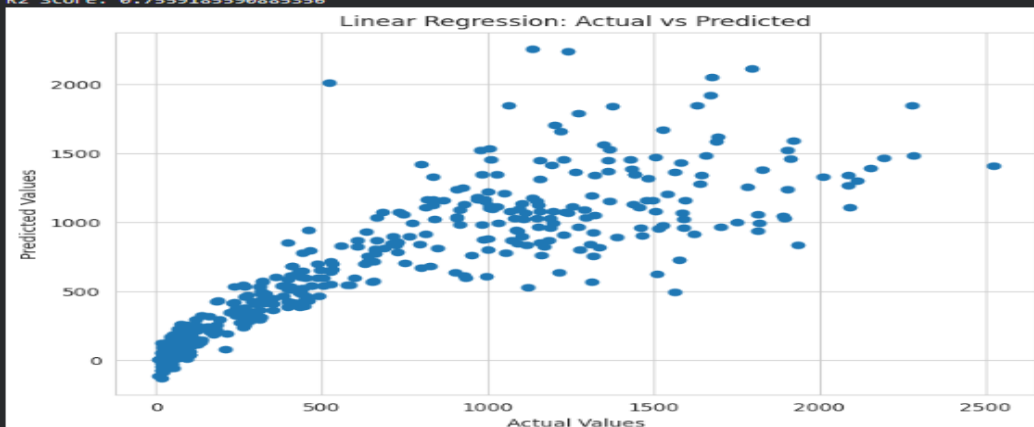


SVR RESULTS
MAE: 385.89066639915785
MSE: 315723.1560550761
RMSE: 561.8924773077819
R2 Score: 0.11205043218706123



Terminal

LINEAR REGRESSION RESULTS
MAE: 191.74594351246074
MSE: 86786.64380551009
RMSE: 294.5957294420781
R2 Score: 0.7559185590885356



The Random Forest Regressor was the best overall performer of all the regression models with the highest R² Score and the lowest errors in prediction. The model was able to model the intricate relationships among customer income, purchasing behavior, and demographic characteristics.

Linear Regression also demonstrated rather good predictive performance, which means that a number of customer-related features had significant linear relationships with spending behavior. Decision Tree Regressor and KNN Regressor produced moderate performance and were able to capture some customer spending patterns effectively.

The Support Vector Regression (SVR) model had the lowest performance of all regression models. The low R^2 Score indicates that the model was not able to generalize well to the data and model customer spending relationships.

The general findings suggest that machine learning methods that rely on ensembles like the Random Forest are more suitable to predict customer spending behavior since they are able to capture non-linear relationships more effectively.

5. Classification Results

Classification models were used to determine whether the customers would react positively to marketing campaigns.

```
print(classification_table)
```

...	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.872768	0.692308	0.268657	0.387097
1	KNN Classifier	0.868304	0.633333	0.283582	0.391753
2	Decision Tree	0.816964	0.394366	0.417910	0.405797
3	Random Forest	0.892857	0.880000	0.328358	0.478261
4	SVM Classifier	0.875000	0.739130	0.253731	0.377778

Random Forest Classifier was the most successful in overall classification with the highest Accuracy, Precision, and F1 Score of all the classification models tested.

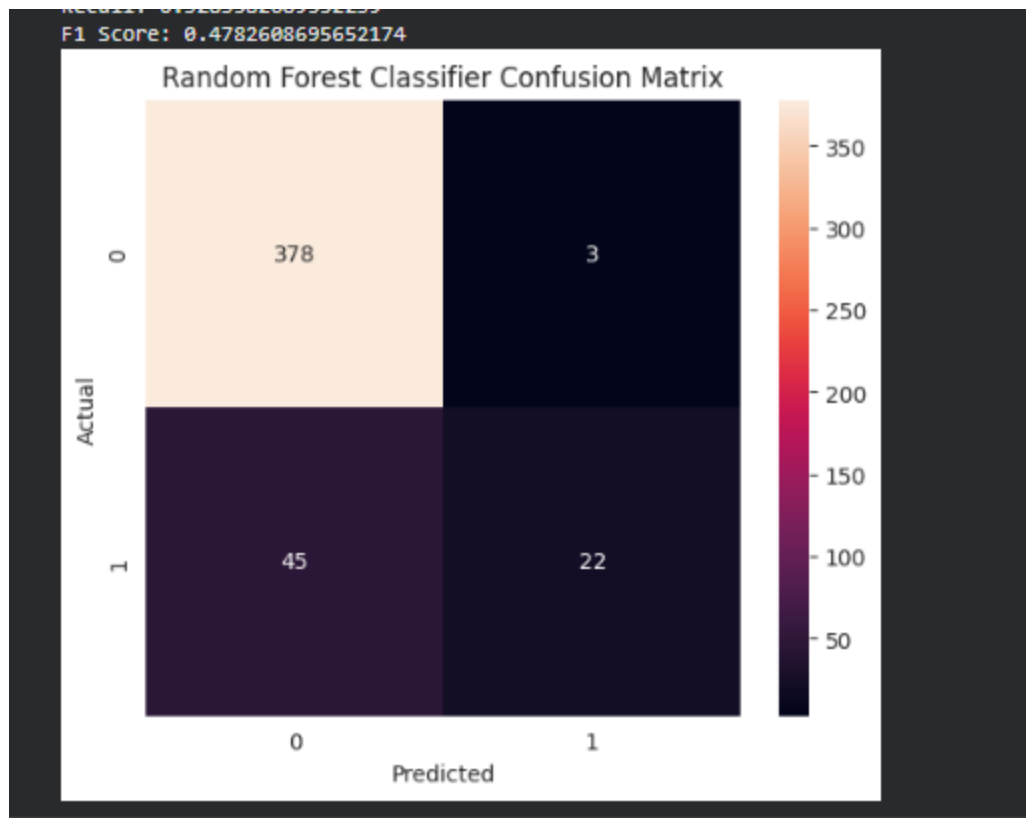
The SVM Classifier and the Logistic Regression models also had fairly good accuracy values. Nevertheless, they have lower recall values, which means that they cannot identify all positive campaign responders.

The Decision Tree Classifier had a better recall than other classification models, i.e. it was capable of recognizing more real positive customer responses.

The results also highlighted the impact of class imbalance in the dataset. Accuracy alone may lead to false conclusions since the majority of customers were not responsive to campaigns. Thus, F1 Score was deemed a more appropriate evaluation measure since it balances Precision and Recall.

The results indicate that ensemble-based algorithms like Random Forest can enhance customer response prediction and assist in targeted marketing strategies.

The best confusion matrix:



6. Discussion

6.1 Optimal Model per Task.

In the case of the regression task, the Random Forest Regressor is the best choice since it has the highest R^2 Score and the lowest prediction errors of all the regression models. The model was able to capture customer spending behavior and yielded the best predictive performance.

In the case of the classification task, the Random Forest Classifier is the best option since it has the highest balance between Accuracy, Precision, Recall, and F1 Score. The model was very effective in the prediction of customer marketing campaign responses.

The outcomes of the project indicate that machine learning methods based on ensembles are very effective in customer analytics and predictive marketing.

6.2 Business Insights

The project produced some valuable business lessons concerning customer behavior and marketing analytics.

The higher income customers tended to show increased spending behavior. There were also significant relationships between customer spending and purchasing activity variables and customer engagement patterns.

The classification analysis showed that it is still difficult to predict campaign responses due to the imbalance in classes. Nevertheless, predictive models can also assist organizations to determine customers who are more inclined to participate in marketing campaigns.

The findings can be used to assist businesses to improve:

- Customer segmentation
- Marketing campaign optimization
- Customer retention strategies
- Revenue forecasting
- Personalized customer targeting

More effective business decision-making and customer relationship management strategies can thus be supported by machine learning models.

6.3 Limitations

The project had several limitations that influenced the project outcomes.

1. First, the classification dataset was imbalanced in terms of classes since fewer customers reacted positively to marketing campaigns. This skew influenced the performance of classification models.
2. Second, sophisticated hyperparameter optimization methods were not used. Certain models may even perform better upon optimization.
3. Third, the data is a particular customer setting and might not be entirely applicable to all business contexts and industries.

Irrespective of these shortcomings, the project was able to illustrate the practical use of machine learning methods in customer analytics and marketing prediction activities.

7. Analysis Questions

Question 1

The best performance was obtained with the Random Forest Regressor with an R^2 Score of 0.8795, which is much better than the worst model, SVR (R^2 : 0.1120). Random Forest model was more effective since it was able to capture more complex non-linear relationships between customer features and spending behavior.

Question 2

Random Forest Classifier had the best F1 Score of all the classification models. As the dataset was imbalanced in classes, F1 Score was more significant than Accuracy as it gave a balanced measure of Precision and Recall.

Question 3

The model results and correlation analysis proved that the variables Income, Total purchases and Recency were significant predictors of customer behavior and marketing campaign responses.

Question 4

False Negatives are the cases when the model predicts that the customer will not respond to a campaign when in fact the customer would have responded positively. This translates to lost marketing opportunities and less effectiveness of campaigns in business terms.

Question 5

Both regression and classification tasks are suggested to be performed with the help of the Random Forest models as they demonstrated the best predictive performance. The models can assist businesses to enhance customer targeting, customer segmentation and marketing optimization strategies.

8. Conclusion

The project was able to use machine learning to analyze customer spending patterns and marketing campaign responses based on the Customer Personality Analysis dataset.

The project featured an entire machine learning workflow such as exploratory data analysis, preprocessing, feature engineering, model training, evaluation and comparison.

The best predictive performance of a customer spending prediction was found in the Random Forest Regressor among all the regression models.

Likewise, the overall classification performance of the Random Forest Classifier was highest in predicting customer campaign responses. The project shows the importance of machine learning in modern organizations as a tool to support customer analytics, business intelligence, targeted marketing, and predictive decision-making. In general, the results prove the usefulness of machine learning in addressing real-life business and marketing issues.

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